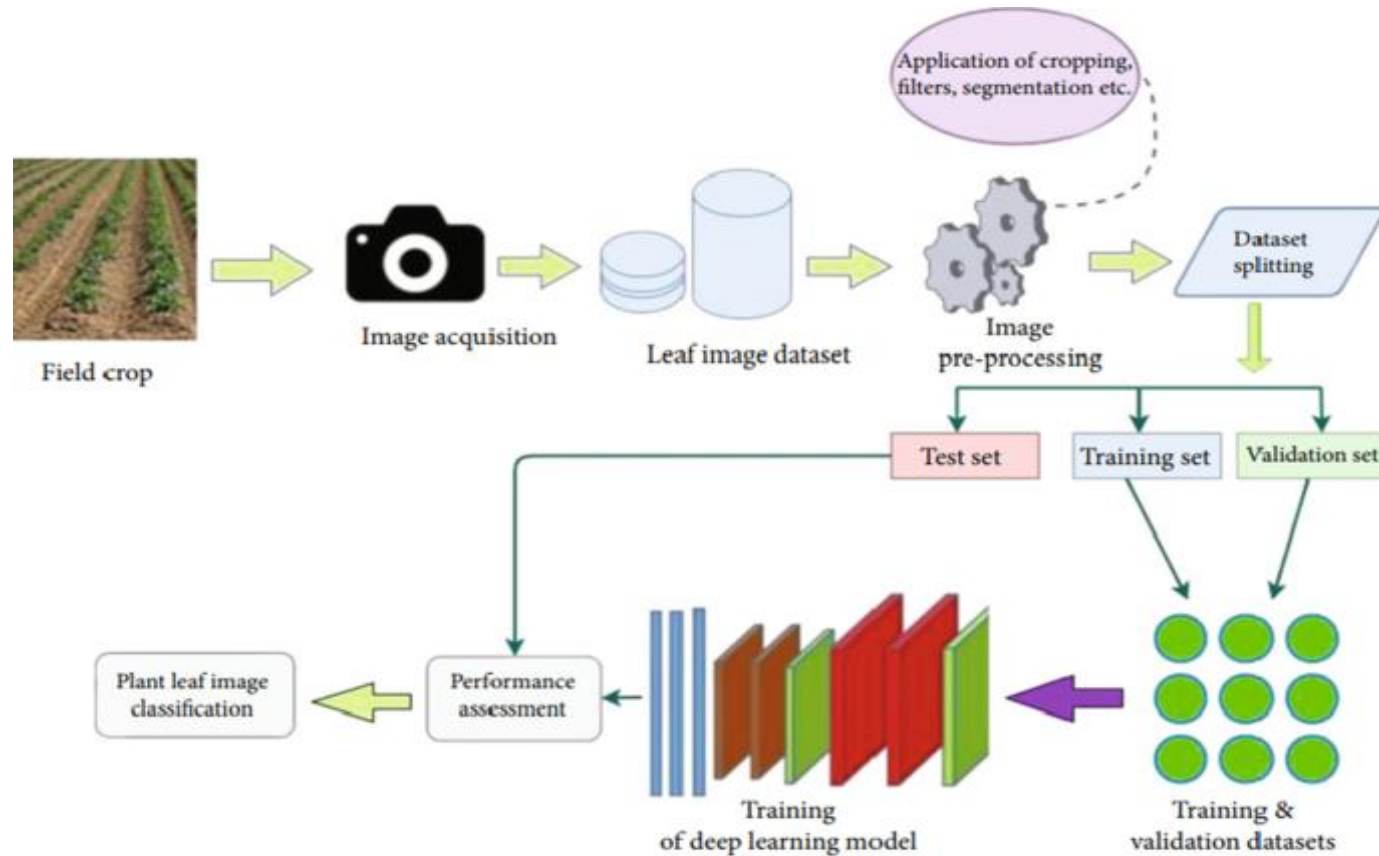


Explainable AI in Agriculture

CNN Interpretability for Crop Disease
Detection

Dr. Maheswari S

CNN Pipeline for Crop Disease Detection



Why Accuracy Is Not Enough

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

No explanation of reasoning

Ignores background bias

Fails in real farm conditions

While this metric tells us **how often the model is correct**, it does not explain:

- Why a particular leaf was classified as diseased
- Whether the model relied on **disease symptoms or background features**
- In real farm conditions, changes in lighting, soil background, or camera angle can cause **unexpected failures**, even if accuracy is high.

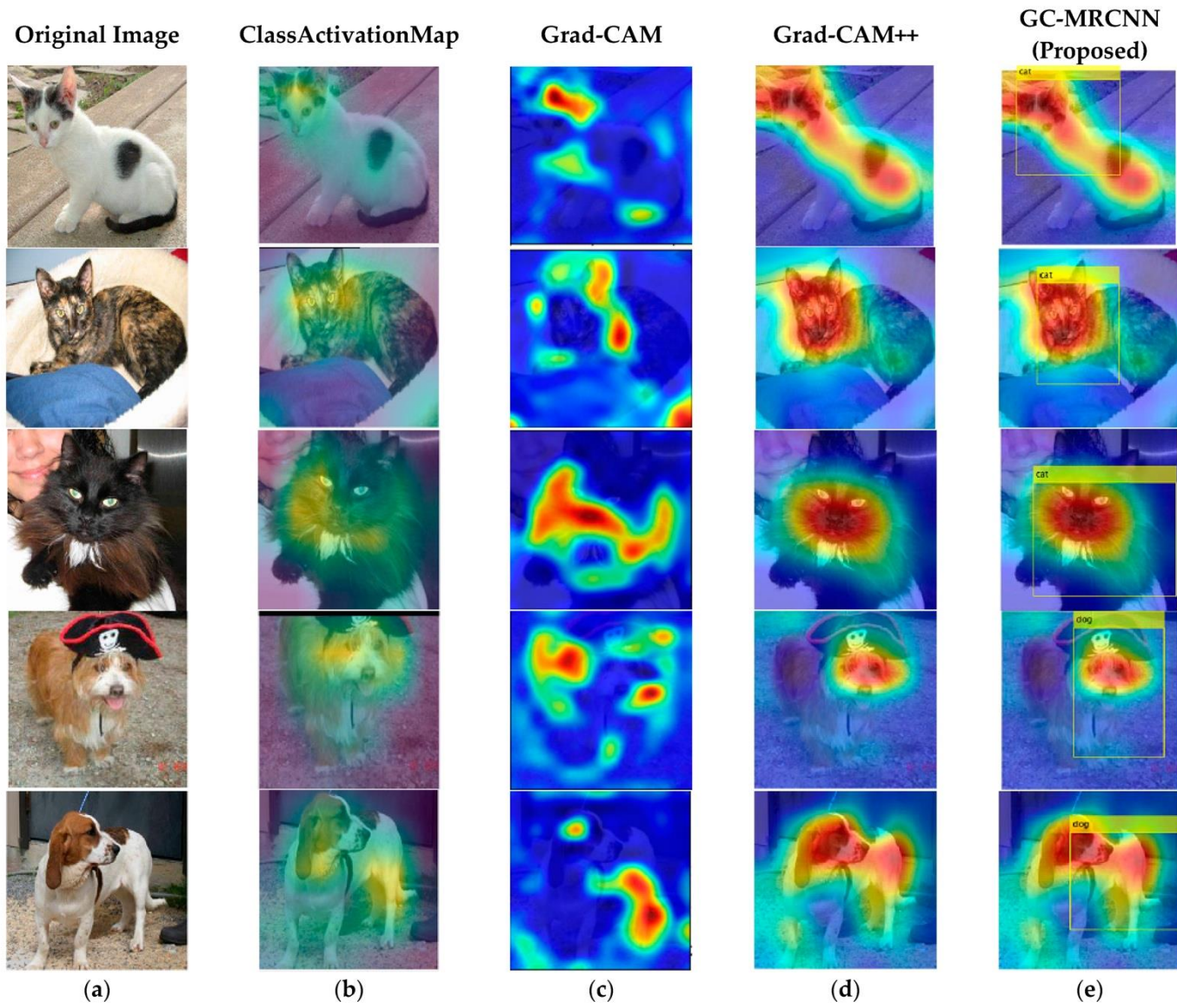
Black-Box Nature of CNNs

- Deep non-linear layers
- Millions of parameters
- No explicit rules

Grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) is an explainable AI technique used to understand why a CNN made a specific prediction by showing which regions of an input image influenced that decision.

Grad-CAM Visualization



Grad-CAM Explanation

Grad-CAM asks:

“Which parts of the image most increased the score for this predicted class?”

It answers this by:

Computing gradients of the predicted class score with respect to the **last convolutional layer**.

Weighting feature maps by how important they are for that class.

Combining them into a heatmap (red = high influence, blue = low).

Overlaying the heatmap on the original image.

Mathematical intuition

For a target class c :

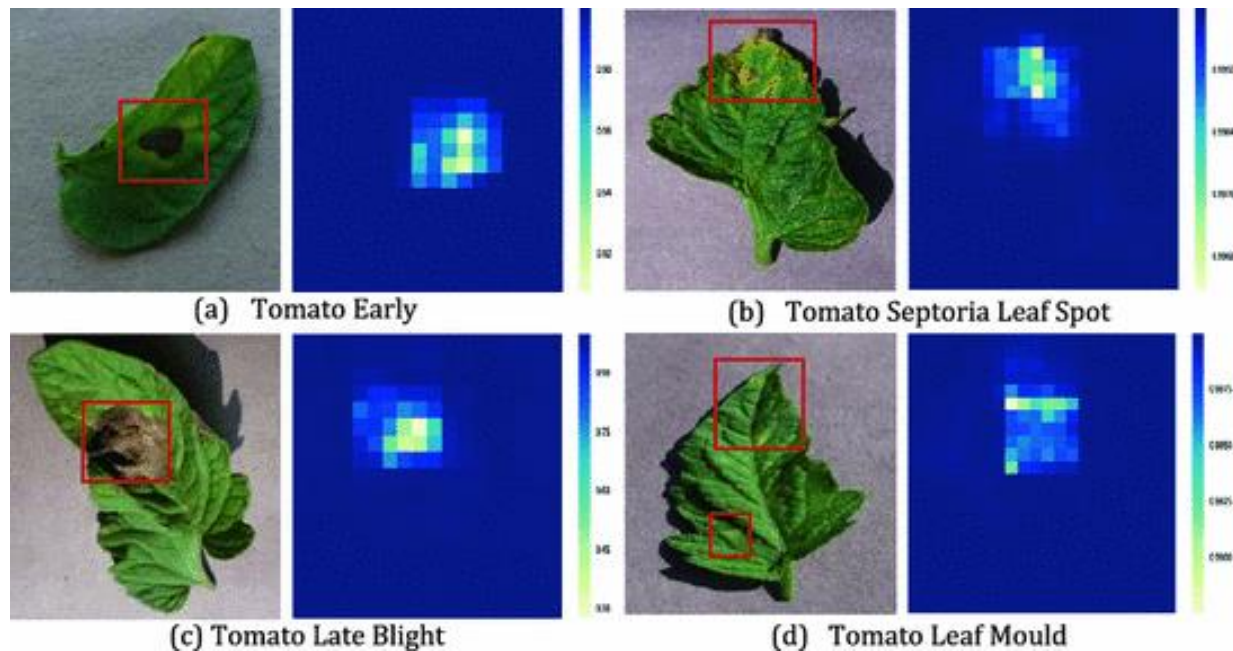
$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$$

Where:

- A^k is the k -th feature map from the last convolutional layer
- y^c is the output score for class c
- α_k^c represents the importance of feature map k
- ReLU ensures only positive contributions are visualized

Saliency Map Visualization (Conceptual)



Original image → Input given to the CNN

Saliency map (heatmap) →

- Bright/white regions: pixels that strongly influence the prediction
- Dark regions: pixels with little influence

The map shows **pixel-level importance**, not object regions

Saliency Maps – Equation

A saliency map highlights **which pixels the model is most sensitive to**, based on gradient magnitude.

$$S = | \partial y / \partial x |$$

- Pixel-wise importance
- Sensitive to noise

Summary

- CNNs are black-box models
- Accuracy alone is insufficient
- Grad-CAM & Saliency Maps provide local explanations
- XAI improves trust in agricultural AI